

Table 3. Construction Stage: New Substation (S/S) + System Strengthening and Augmentation Works of Existing Substations (S/S)						
Key Activities / Potential Impacts & Issues during Construction Stage	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁴	Monitoring Parameter	Method of Measurement	Frequency
	E.2	Prohibit the use of chemical dust suppressants	new S/S + existing S/S	Chemical dust suppressants avoided	Ocular inspection / spot-checks	Frequent spot checks
	E.3	Dust prone materials / stockpiles to be stored in enclosures and/or covered with tarp	new S/S + existing S/S	Materials / Stockpiles storage area	Ocular inspection / spot-checks	Frequent spot checks
	E.4	Periodic maintenance of heavy equipment, machinery, transformers, construction vehicles to minimise air pollution, noise, and reduce OHS risks	area of influence	Heavy equipment, machinery, transformers, construction vehicles	Ocular inspection / Maintenance log-book	Weekly
	E.5	Control vehicle speed to ≤ 8 km/h in unpaved areas including unpaved approach roads; Post the speed limit sign in the project work areas	area of influence	Speed limit signage	Ocular inspection / spot checks	Frequent spot checks
	E.6	Vehicles transporting materials that generate dust will be covered with tarps	area of influence	Provision of tarps on vehicles	Ocular inspection / spot checks	Frequent spot checks
	E.7	Undertake air quality monitoring using handheld / portable air monitoring devices at select locations for OHS and CHS purposes	within new S/S + existing S/S, and sensitive receptors immediately downwind, if any	PM10, PM2.5, NOx, SOx, CO	Monitoring conducted using hand held / portable air monitoring devices	Monthly during peak construction + once after completion of construction
	E.8	Comply with relevant National standards on air quality / IFC (WBG) Environmental, Health, and Safety (EHS) Guidelines on Air and Ambient Air Quality, Electric Transmission and Distribution ²⁷	new S/S + existing S/S	Compliance with National standards and/or IFC (WBG) Guidelines, whichever is stringent	Record and maintain log of monitoring / incidences of non-compliance and rectification	Monthly during construction + once after completion of construction

²⁷ <https://www.ifc.org/content/dam/ifc/doc/2000/2007-general-ehs-guidelines-air-emissions-and-ambient-air-quality-en.pdf>; <https://www.ifc.org/content/dam/ifc/doc/2000/2007-electric-transmission-distribution-ehs-guidelines-en.pdf>

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Key Activities / Potential Impacts & Issues during Construction Stage	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁴	Monitoring Parameter	Method of Measurement	Frequency
	E.9	Implement S-EMP: Dust Control	new S/S + existing S/S	S-EMP implementation	Log-book	Daily
		Noise				
Increase in noise due to movement of heavy equipment, machinery, construction vehicles, civil works (levelling, grading, back-filling, etc)	F.1	Construction activities utilizing heavy equipment, machinery work will be restricted between 6 AM – 7 PM	new S/S + existing S/S	Construction timings	Log-book	Daily
	F.2	Any night time works only after receiving due permission from relevant authority and advance consultations with nearby households / settlements – (also refer to Measures listed under Community Health and Safety, Table 3)	new S/S + existing S/S	Copy of permission night time works	Permission on file	Once prior to scheduled night time project activity
	F.3	Advance warning and sharing of Construction Implementation Schedule including night time works to area Gram Panchayat, nearby households / settlements ²⁸	new S/S + existing S/S	Consultations, Information dissemination via leaflets	Consultations log-book; leaflets distributed	Once prior to scheduled project activity
	F.4	Ensure all generators, heavy equipment, machinery, etc are sound insulated and placed in enclosures to minimise ambient noise levels; situated away from construction (workers) camps	new S/S + existing S/S	Guaranteed noise level of corresponding heavy equipment, machinery	Equipment/ machinery specifications	Once prior to scheduled project activity

²⁸ Measures in F.3, Table 3 to be undertaken in coordination with the PIU

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Key Activities / Potential Impacts & Issues during Construction Stage	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁴	Monitoring Parameter	Method of Measurement	Frequency
	F.5	Periodic maintenance of heavy equipment, machinery, transformers, construction vehicles to minimise emissions and noise, and reduce OHS risks	area of influence	Construction vehicles, machinery, equipment	Ocular inspection / Maintenance log-book	Weekly
	F.6	Undertake noise quality monitoring using handheld / portable noise monitoring devices at select locations for OHS / CHS purposes	new S/S + existing S/S and sensitive receptors downstream, if any	Day time / night time noise	Monitoring conducted using hand held / portable devices	Monthly during peak construction + once during commissioning of new S/S
	F.7	Comply with relevant National standards on noise, / IFC (WBG) Environmental, Health, and Safety (EHS) Guidelines on Noise, Electric Transmission and Distribution ²⁹	new S/S + existing S/S	Compliance with National standards and/or IFC (WBG) Guidelines, whichever is stringent	Record and maintain log of monitoring / incidences of non-compliance and rectification	Monthly during construction + once during commissioning of new S/S
	F.8	Implement S-EMP: Noise Control	new S/S + existing S/S	S-EMP implementation	Log-book	Daily
		Polluting Materials and Pollution Management				
Pollution due to oil, lubricant, fuel spillage	G.1	Allocate and maintain a temporary secured and/or controlled area for storage and handling of hazardous materials with a containment tray or bunds; away from direct sunlight and rain	new S/S + existing S/S	Temporary secure and/or controlled area set up in place	Ocular inspection / spot checks	Weekly

²⁹ <https://www.ifc.org/content/dam/ifc/doc/2000/2007-general-ehs-guidelines-noise-en.pdf>; <https://www.ifc.org/content/dam/ifc/doc/2000/2007-electric-transmission-distribution-ehs-guidelines-en.pdf>

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Key Activities / Potential Impacts & Issues during Construction Stage	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁴	Monitoring Parameter	Method of Measurement	Frequency
	G.2	Oil contaminated water from containment trays or bunds to be removed by means of a manually controlled positive lift pump, or oil absorbent pads or other means -- all to be agreed in advance with licensed vendors	new S/S + existing S/S	Removal of oil contaminated water mechanism is in place	Ocular inspection / Maintenance log-book	Weekly or as required during project activity
	G.3	Oil containment system as per design provided at transformer pad	new S/S + existing S/S	Power transformer pad built and maintained	Ocular inspection	Once
Inappropriate handling, storage, transport and disposal	G.4	Collection of oily rags and waste oil in proper containment units	new S/S + existing S/S	Containment units provided in secured / controlled area	Ocular inspection	Weekly or as required during project activity
	G.5	Record of incidents, spills to be maintained at work sites	new S/S + existing S/S	Incidents, spills	Ocular inspection / log-book	Daily
Quality of transformer oil	G.6	Material Safety Data Sheet (MSDS) – compliance to IS: 1866: Code of Practice for Electrical Maintenance and Supervision of Mineral Insulating Oil in Equipment	new S/S + existing S/S transformers	Record of MSDS	Log-book, record of MSDS onsite, certification checks	Weekly or as required during project activity
Materials Safety Data Sheet (MSDS)	G.7	Delivery and acceptance of all hazardous materials, equipment to be accompanied by a Materials Safety Data Sheet (MSDS) and /or to be certified that it is Polychlorinated Bi-phenyl (PCB), Chlorofluorocarbons (CFCs) including halon, free	new S/S + existing S/S	Record of equipment and corresponding PCB, CFCs, Halon free certificate; Record of MSDS	Log-book, record of MSDS onsite, certification checks	Weekly or as required during project activity
Use of Licensed Vendors	G.8	MSEDCL / licensed vendors to collect used transformers from existing S/S for	existing S/S	Licensed vendor list	Log-book with record of trip, location and distance	As required during project activity

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Key Activities / Potential Impacts & Issues during Construction Stage	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁴	Monitoring Parameter	Method of Measurement	Frequency
		repair, re-use ³⁰				
	G.9	Record of materials with corresponding location and distance for final storage / disposal / re-use from work sites with date, location maps and number of trips to be maintained onsite	new S/S + existing S/S	Materials inventory	Ocular inspection / log-book	Daily
	G.10	Comply with relevant National standards in hazardous and polluting materials / IFC (WBG) Environmental, Health, and Safety (EHS) Guidelines for Hazardous Material Management, Waste Management, Electric Transmission and Distribution ³¹	new S/S + existing S/S	Compliance with National standards and/or IFC (WBG) Guidelines, whichever is stringent	Record and maintain log of monitoring / incidences of non-compliance and rectification	Monthly
	G.11	Implement S-EMP: Hazardous Materials Control Plan (including spills response and clean up)	new S/S + existing S/S	S-EMP implementation	Log-book	Daily
		Waste Management (all waste streams)				
Improper waste management, handling and disposal	H.1	Allocate and maintain space within work sites for a covered, designated area for collection, storage, sorting and recycling; at least 10 m distance from water sources / bodies, drainage, bunds, etc	new S/S	Sorting area established; 10 m distance maintained	Ocular inspection / spot checks	Weekly
	H.2	Prohibit waste burning	new S/S + existing S/S	Waste management	Ocular inspection / spot checks	Daily

³⁰ Note: PCB-containing equipment can be used until December 31, 2025 or the equipment's certified lifetime, whichever is earlier. The removed equipment shall be properly maintained to prevent PCBs from leaking or being released into the environment. PCB-contaminated equipment is regulated as hazardous waste under the Hazardous Wastes (Management, Handling and Transboundary Movement) Rules, 2008.

³¹ <https://www.ifc.org/content/dam/ifc/doc/2000/2007-general-ehs-guidelines-hazardous-materials-management-en.pdf>; <https://www.ifc.org/content/dam/ifc/doc/2000/2007-general-ehs-guidelines-waste-management-en.pdf>; <https://www.ifc.org/content/dam/ifc/doc/2000/2007-electric-transmission-distribution-ehs-guidelines-en.pdf>

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Key Activities / Potential Impacts & Issues during Construction Stage	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁴	Monitoring Parameter	Method of Measurement	Frequency
	H.3	Provide multiple waste containers	new S/S + existing S/S	Waste management	Ocular inspection	Weekly
	H.4	Collection of waste by licensed vendors	new S/S + existing S/S	Licensed vendor list	Log-book with record of trip, location and distance	As required during project activity
	H.5	Transport of recyclables to dedicated MSEDCL yards / filter units for re-use, repair; resale or auction to government authorized vendors only	new S/S + existing S/S	Licensed vendor list for auction	Inventory / Log-book with record of trip, location and distance	As required during project activity
	H.6	Ensure proper removal and disposal of any significant residual materials, wastes that remain onsite after completion of civil works; no final disposal on site; No disposal on farmland	new S/S + existing S/S	Waste management	Record of waste disposal sites with corresponding distance from work site, date, location map and number of trips to be maintained at work site	Once after completion of construction
	H.7	Comply with relevant National standards on waste management / IFC (WBG) Environmental, Health, and Safety Environmental Guidelines on (General and Hazardous) Waste Management, Electric Transmission and Distribution) ³²	new S/S + existing S/S	Compliance with National standards and/or IFC (WBG) guidelines, whichever is stringent	Record and maintain log of monitoring / incidences of non-compliance and rectification	Monthly
	H.8	Implement S-EMP: Waste Management (all waste streams)	new S/S + existing S/S	S-EMP implementation	Log-book	Daily

³² <https://www.ifc.org/content/dam/ifc/doc/2000/2007-general-ehs-guidelines-waste-management-en.pdf>; <https://www.ifc.org/content/dam/ifc/doc/2000/2007-electric-transmission-distribution-ehs-guidelines-en.pdf>

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Key Activities / Potential Impacts & Issues during Construction Stage	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁴	Monitoring Parameter	Method of Measurement	Frequency
		Occupational Health and Safety (OHS)				
Provision of construction (workers) camps	I.1	Ensure adequate workers accommodation in line with IFC (WBG) Guidance Note / Workers Accommodation and measures in A.3 - A.5, Table 2)	new S/S	Provision of accommodation and adequate services	Ocular inspection / spot checks	Once
Unhygienic conditions at construction (workers) camps	I.2	Construction (workers) camps to have adequate facilities for temporary stay such as electricity, portable water, portable toilets, waste collection areas /bins, first aid kit, etc	new S/S	Provision of adequate water and sanitation facilities, electricity connection	Ocular inspection / spot-checks	Weekly
Exposure to hazards for workers working on concrete mixing, cement, etc	I.3	Concrete, cement mixing process for S/S building, asphalt mixing for access road construction located at least 10 m downwind of nearest settlement and/or construction (workers) camps	new S/S + existing S/S	10 m distance maintained	Ocular inspection	Once
Exposure to hazards for workers working on concrete mixing, cement, etc	I.4	Strictly enforce the use of PPE as such goggles, rubber gloves, sleeves, blankets, noise reducing mufflers, head lamps, high visibility safety vests with reflective striping (e.g., for night time works if any), respirators (e.g., working on concrete / cement, etc.)	new S/S + existing S/S	PPE provided	Inventory and, PPE check / ocular inspection	Weekly
	I.5	Provide communication devices to contractor supervisors / C-EHS focal point person	new S/S + existing S/S	Communication devices provided	Inventory check	Once
Electrical hazards, Working at heights, etc	I.6	Assess safety risks and safety protocols as per MSEDCL requirements and	new S/S + existing S/S	Safety risks / protocols assessed for OHS	Safety check	Weekly

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Key Activities / Potential Impacts & Issues during Construction Stage	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁴	Monitoring Parameter	Method of Measurement	Frequency
		implement at work sites ³³				
	I.7	When testing electrical equipment, all unrelated workers to leave the 'flagged or danger zone' and comply with the relevant MSEDCL electrical safety standards	new S/S + existing S/S	Safety risks / protocols assessed for OHS	Safety check	During testing of electrical equipment
	I.8	Provide and maintain signage as per MSEDCL standards for electrical safety	new S/S + existing S/S	Adequate signage provided	Ocular inspection	Once
Fire, explosion and other incidences / accidents	I.9	Periodic training of workers for specific type of work engagement, e.g., OHS, hazardous materials handling, waste disposal, working at heights, electrical works, safety protocols, and Emergency Response Plans (ERPs) including fire and explosion	new S/S + existing S/S	OHS training	OHS training log-book	Once every 3 months
	I.10	Ensure provision of fire hydrants, sand buckets at work sites; regularly checked for expiration date, refilled, replaced as required (prior to the expiration date); Any wet sand (if any) in the bucket should be replaced with dry sand	new S/S + existing S/S	Provision of fire hydrants	Ocular inspection / spot-checks	Monthly
Health incidences, workers safety	I.11	Coordinate with nearest hospital / local clinic in case of emergencies and/or work accidents	new S/S + existing S/S	Clinic / hospital list	Log-book	Once

³³ [MSEDCL Training and Safety](#)

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Key Activities / Potential Impacts & Issues during Construction Stage	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁴	Monitoring Parameter	Method of Measurement	Frequency
	I.12	Set up first aid kit within work sites	new S/S + existing S/S	Provision of first aid kits	Inventory check / spot checks	Once
	I.13	Provide periodic health check-ups (health & fitness) to workers	new S/S + existing S/S	Health check-ups	Record of health assessments	Once every 3 months
	I.14	Undergo orientation in health and safety risks due to transmittable diseases (HIV, AIDS, Covid-19)	new S/S + existing S/S	Worker's orientation in OHS	Record of orientation	Once every 3 months
	I.15	Maintain a record of health assessments, orientation, incidents, spills, accidents, near miss, fatalities in a log-book at work sites	new S/S + existing S/S	Incidents, spills, accidents, near miss, fatalities	Record of incidences in log-book	Once every 3 months
	I.16	Comply with relevant National laws and regulations on Occupational Health and Safety / IFC (WBG) Environmental, Health and Safety (EHS) Guidelines on Occupational Health and Safety ³⁴	new S/S + existing S/S	Compliance with National laws and regulations and/or IFC (WBG) Guidelines, whichever is stringent	Record and maintain log of monitoring / incidences of non-compliance and rectification	Monthly
	I.17	Implement S-EMP: Occupational Health and Safety Plan	new S/S + existing S/S	S-EMP implementation	Log-book	Daily
	I.18	Implement S-EMP: Emergency Response Plan	new S/S + existing S/S	S-EMP implementation	Log-book	Daily
		Community Health and Safety (CHS)³⁵				

³⁴ <https://www.ifc.org/content/dam/ifc/doc/2000/2007-general-ehs-guidelines-occupational-health-and-safety-en.pdf>

³⁵ Measures listed under Community Health and Safety J.1 through J.10, Table 3 to be undertaken in coordination with the PIU

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Key Activities / Potential Impacts & Issues during Construction Stage	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁴	Monitoring Parameter	Method of Measurement	Frequency
Excessive disturbances to communities due to civil works, increase in vehicular / equipment / machinery traffic	J.1	Issue advance warning to households / settlements viz area Gram Panchayat including erection of signage of planned access routes, detours, work timing (including any scheduled night time works), duration at appropriate places along road and at work sites; deploy workers to direct traffic in case of lane / road closure	Households / settlements in and around new S/S + existing S/S	Issuance of advance notice, danger and warning signs, workers deployed corresponding to the construction implementation schedule	Advance warning / information disseminated	As per construction implementation schedule
Disruption to utilities	J.2	Confirm with relevant local agency for existing utilities (UG pipes, telecom) prior to commencing civil / augmentation works	new S/S	Damage, disadvantage to local communities, businesses, persons	Ocular inspection / spot-checks	As per construction implementation schedule
Lack of access to nearby areas	J.3	Maintain access to grazing sites, adjacent properties, access lanes, roads or provide nearest alternate route marked on map	new S/S + existing S/S	Access maintained	Ocular inspection / spot-checks	Duration construction
Accidents due to access to work sites	J.4	Ensure all work sites are secure during peak construction; discourage access by public through appropriate fencing, barricades, warning cones (e.g., around work sites, temp access tracks, etc), signage, sufficient lighting	new S/S	Perimeter fencing installed, lighting installed, danger and warning signs for safety of community, security personnel deployed	Ocular inspection / spot-checks	Weekly
Damage to project assets	J.5	Assign security personnel, flag persons to prevent access to work sites, accidents, trespassing, vandalism of equipment and/or pilferage of lines / cables	new S/S + existing S/S	Security personnel assigned to work sites	Attendance records of security personnel maintained	Daily

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Key Activities / Potential Impacts & Issues during Construction Stage	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁴	Monitoring Parameter	Method of Measurement	Frequency
CHS issues	J.6	Conduct meaningful consultations with households/ settlements to keep them informed on project implementation schedule (and changes if any), in particular those that may result in disruption with respect to area access, vehicular movement, utilities, noisy / dust generating activities, night time works, etc; distribute project information leaflet in local language including Posters on Awareness of Electrical Safety ³⁶	area of influence	Information disclosure, participation and consultation including distribution of project information leaflet, project GRM, etc	Consultation log-book	Monthly during construction + once at completion of construction
	J.7	Inform and consult with the community on project GRM	area of influence	Grievances received, resolution if any	Maintain a record of grievances raised in a GRM log-book at work site, records to be maintained in line with project GRM	Monthly during construction + once at completion of construction
	J.8	Comply with National laws and regulations on Community Health and Safety / IFC (WBG) Environmental, Health, and Safety (EHS) Guidelines on Community Health and Safety ³⁷	area of influence	Compliance with National laws and regulations and/or IFC (WBG) Guidelines, whichever is stringent	Record and maintain log of monitoring / incidences of non-compliance and rectification	Monthly
	J.9	Implement S-EMP: Community Health and Safety Plan	area of influence	S-EMP implementation	Log-book	Daily

³⁶ [Posters on Awareness of Electrical Safety](#)

³⁷ <https://www.ifc.org/content/dam/ifc/doc/2000/2007-general-ehs-guidelines-community-health-and-safety-en.pdf>

Table 3. Construction Stage: New Substation (S/S) + System Strengthening and Augmentation Works of Existing Substations (S/S)						
Key Activities / Potential Impacts & Issues during Construction Stage	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁴	Monitoring Parameter	Method of Measurement	Frequency
	J.10	Any damage or hindrance, disadvantage to local communities, businesses, persons etc., caused by the premature removal or insufficient replacement of public utilities is subject to full compensation, at full liability of the Contractor who caused the problem	area of influence	Part of contract clause	Relevant local agency consulted; utilities marked on site maps; records maintained	Once after completion of construction
		Physical Cultural Resources (PCRs)³⁸				
Chance Find of PCRs	K.1	Implement S-EMP: Chance Find Procedures for PCRs as per measures K.1.1 through K.1.4, Table 3 below:	in and around new S/S + existing S/S + wider area of influence	PCRs	Ocular inspection / spot-checks	Once
	K.1.1	Civil works suspended immediately if PCRs encountered and Contractor supervisor (C-EHS) to be informed				
	K.1.2	Prohibit destroying, damaging, defacing or concealing PCRs				
	K.1.3	Inform and consult with the district office of Culture and Fine Arts official				
	K.1.4	Civil works to resume only after thorough investigation and permission from district office of Culture and Fine Arts				

³⁸ Measures in K.1, Table 3 to be undertaken in coordination with the PIU

Table 4. Construction Stage: New Distribution Lines (DT) + System Strengthening and Reconductoring of Existing Distribution Lines (DT)						
Key Activities / Potential Impacts & Issues	No.	Mitigation Measures and Safeguards	Monitoring Location ³⁹	Monitoring Parameter	Method of Measurement	Frequency
		Local Ecology				
Impact on habitat, species due to site preparation + civil works	A.1	Bird / bast roosting sites, if any found, will be marked and not disturbed at work sites	new DT + existing DT ROW	Roosting sites marked	Ocular inspection	Once before start of site preparation + construction
	A.2	Schedule activities in dry season and avoid sensitive periods (breeding periods for aves)	area of influence	Breeding period of aves ascertained	Project work scheduled for dry season and/or avoid sensitive periods of breeding for aves schedule	Once before start of site preparation + construction
		Land and Vegetation				
Impact on land and vegetation due to site preparation + civil works	B.1	Clear demarcation of pole footings along selected ROW, no encroachment outside demarcated ROW	pole footings along new DT + existing DT ROW	Clear demarcation of ROW, pole footings	Ocular inspection / spot-checks	Once before start of site preparation + construction
	B.2	Vegetation clearance strictly restricted to ROW, temp access tracks - in accordance with safety clearance requirements; use existing approach roads to ROW as much as possible	new DT + existing DT ROW, temp access tracks	Land /vegetation clearance	Ocular inspection / spot-checks	During site preparation + construction
Impact on VERs	B.3	Confirm and verify VERs are avoided; tree cutting avoided and/or due permission from competent authority obtained and on file	new DT + existing DT ROW, temp access tracks	VERs; tree type + tree height + tree count / compensation ratio and offset options	VERs avoided; Tree cutting permission on file; records maintained	Once before start of site preparation + construction

³⁹ S/S = substation; DT = distribution line; ROW = Right of Way; Project defined area of influence = new substation or existing substation and buffer of 50 m around the substation, new distribution line or existing distribution line and a buffer of 25 m on each side of the right of way center-line

Table 4. Construction Stage: New Distribution Lines (DT) + System Strengthening and Reconductoring of Existing Distribution Lines (DT)						
Key Activities / Potential Impacts & Issues	No.	Mitigation Measures and Safeguards	Monitoring Location ³⁹	Monitoring Parameter	Method of Measurement	Frequency
	B.4	Tree looping / pruning will be done by trained arborist to maintain minimum sag	new DT + existing DT ROW	Land /vegetation clearance	Ocular inspection / spot-checks	During site preparation
Use of pesticides and herbicides	B.5	Use of pesticides or herbicides for land / vegetation clearance is prohibited; employ use of manual or machine clearance	new DT + existing DT ROW, temp access tracks	Vegetation clearance method by manual or machine clearing	Ocular inspection / spot checks	During site preparation
Materials Usage	B.6	Maximize the re-use of spoils for backfilling at base of DT poles; ensure usage results in no interference to natural drainage pattern	pole footings along new DT + existing DT ROW	Materials sources / stocks	Ocular inspection / spot-checks / Materials log-book	Weekly
Materials / Stockpiling	B.7	Stockpiles (poles, cable reels, wires, etc) to be placed within nearest MSEDCL warehouses / filter units or Contractor owned warehouse, and not along DT ROW except prior to erection or installation. Ensure access to area is not blocked; ensure at least 10 m distance from a water source is maintained)	new DT + existing DT ROW	Stockpiles storage area within nearest MSEDCL warehouses / filter units or Contractor owned warehouse; Area access maintained, 10 m distance from water source maintained	Ocular inspection / spot-checks / Materials log-book	Weekly
	B.8	Implement S-EMP: Materials Management Plan (including warehouses / storage)	new DT + existing DT	S-EMP implementation	Materials Log-book	Weekly
	B.9	Disposal of spoils (if any) on paddy fields, farmland, or within 50 m of a water source is prohibited	area of influence	Farmland or water source in the wider area of influence	Ocular inspection / spot checks / log-book with number of trips and spoil disposal location	Weekly during construction or as required during project activity
	B.10	Ensure vehicles are covered during transportation of stockpiles, spoils to avoid spillage	area of influence	Provision for vehicles cover	Ocular inspection	During transportation

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Key Activities / Potential Impacts & Issues	No.	Mitigation Measures and Safeguards	Monitoring Location ³⁹	Monitoring Parameter	Method of Measurement	Frequency
Soil erosion	B.11	During excavation of pole base, slopes will be maintained at safe angles to avoid risk of collapse	pole footings along new DT + existing DT ROW	Slope angles	Ocular inspection	Once at time of excavation of pole foundations
	B.12	Immediately cover and stabilize excavation and disturbed ground after DT pole erection, grading for soil stabilization, drainage and/or other erosion prone areas, ensure closing of any excavations or trenches	new DT + existing DT pole erection	Construction (workers) camps	Ocular inspection / spot checks	Immediately after completion of construction
Presence of construction (workers) camps	B.13	Construction (workers) camps maintained in demarcated areas as per requirements listed in Table 2, Items A.3, A.4 and A.5 and in accordance with IFC (WBG) Guidance Note / Workers Accommodation	new DT + existing DT	Construction (workers) camps	Ocular inspection / spot checks	Weekly
Land restoration	B.14	Re-seeding along any backfilled areas with native vegetation / soil binding grass for restoring habitat (and soil stabilization) at least within 30 days of completion of civil works; for activities carried out during periods of hot or extreme weather, ensure adequate after care to maximize survival	new DT + existing DT ROW	Re-seeded areas / number of native species planted	Ocular inspection / log-book	Once within 30 days of completion of construction
	B.15	Restore temporary access tracks and/or temp materials storage areas to pre-constr conditions at least within 30 days of completion of civil works	new DT + existing DT ROW; erection of poles and stringing of conductors	Restored temp access tracks / materials storage materials area	Ocular inspection	Once within 30 days of completion of construction

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Key Activities / Potential Impacts & Issues	No.	Mitigation Measures and Safeguards	Monitoring Location ³⁹	Monitoring Parameter	Method of Measurement	Frequency
	B.16	Implement S-EMP: Site Rehabilitation and Clean-up plan	new DT + existing DT ROW	S-EMP implementation	Log-book	Once after completion of construction
Damage to infrastructure	B.17	Any damage to areas or infrastructure outside the agreed work sites (corridor of impact assessed as per social safeguards due diligence) to be restored to pre-construction conditions and will be subject to compensation at contractor cost and through written agreement with the land owner / relevant agency, as applicable ⁴⁰	new DT + existing DT ROW; erection of poles and stringing of conductors	Part of the contract clause	Ocular inspection	Once after completion of construction
		Stormwater, Drainage Management				
Disturbance to natural drainage pattern due to civil works	C.1	Backfilling, use / re-use of materials or excess spoils, placement of stockpiles to take due cognizance of, maintain the natural drainage pattern and causes no disturbance to nearby existing natural water sources / bodies, drainage, bunds, etc; Keep loose soil material and stockpiles out of drains and flow-lines)	new DT + existing DT ROW	Natural drainage pattern maintained; drains and flow-lines cleaned and maintained	Ocular inspection / spot checks	Weekly during construction + once after completion of construction
		Water Resources, Water Quality				
Impact on water resources / groundwater sustainability	D.1	Water usage for civil works as per stipulated quantity or volume sanctioned by the local water supply service provider	new DT + existing DT	Memorandum of Understanding (MOU) between contractor and service provider	Project specific commitment letter on file	Once

⁴⁰ Measures in B.16, Table 4 to be undertaken in coordination with PIU

Table 4. Construction Stage: New Distribution Lines (DT) + System Strengthening and Reconductoring of Existing Distribution Lines (DT)						
Key Activities / Potential Impacts & Issues	No.	Mitigation Measures and Safeguards	Monitoring Location ³⁹	Monitoring Parameter	Method of Measurement	Frequency
Materials / Stockpiles	D.2	Construction (workers) camps, temporary placement of stockpiles along ROW shall be at least 10 m distance from water sources, drainage; Keep loose soil material and stockpiles out of drains and flow-lines	new DT + existing DT ROW	10 m distance maintained; drains and flow-lines cleaned and maintained	Ocular inspection / spot checks	Weekly
Siltation / contamination of water sources / water bodies due to spillage of slurry, wastes, constr debris, oil, fuel, etc	D.3	Provision of temporary adequate receiving body (e.g., bunds, tanks) in and around pole foundation, as deemed applicable, without causing onsite / offsite adverse impacts from construction run-off and/or water logging; After collection, process water and muddy runoff and if necessary, employ flocculants such as PAM (Polyacrylamide) to facilitate sedimentation or filter to remove silts)	new DT + existing DT	Provision of an adequate receiving body (bunds, tanks) for construction run-off onsite	Ocular inspection	Once
	D.4	Avoidance of waterways in site selection for pole placement	new DT + existing DT	Placement of tower footings in water bodies avoided as per design	Ocular inspection	Once
	D.5	Provision and maintenance for temp sanitation (portable toilets) facilities for workers (separate for genders) in Construction (workers) camps	construction (workers) camps	Provision of temp sanitary facilities	Ocular inspection / spot-checks / Maintenance log-book	Daily
	D.6	Collection of sanitation waste by licensed vendor, to be transported by specialized trucks with units to nearest waste disposal facilities	construction (workers) camp	Memorandum of Understanding (MOU) between contractor and licensed vendor	Project specific commitment letter on file; record of number of trips maintained	Weekly during construction or as required during project activity
	D.7	Conduct surface water quality monitoring	any natural receiving water source/ body in close proximity to	Surface water quality monitoring conducted	Sampling and chemical analysis	Quarterly during construction + once after completion construction

Table 4. Construction Stage: New Distribution Lines (DT) + System Strengthening and Reconductoring of Existing Distribution Lines (DT)						
Key Activities / Potential Impacts & Issues	No.	Mitigation Measures and Safeguards	Monitoring Location ³⁹	Monitoring Parameter	Method of Measurement	Frequency
			DT ROW			
	D.8	Comply with relevant National standards on water quality / IFC (WBG) Environmental, Health, and Safety (EHS) Guidelines on Wastewater and Ambient Water Quality, Electric Transmission and Distribution ⁴¹	new DT + existing DT	Compliance with National standards or IFC (WBG) Guidelines, whichever is stringent	Record and maintain log of monitoring / incidences of non-compliance and rectification	Quarterly during construction + once after completion construction
	D.9	Implement S-EMP: Construction Wastewater	new DT + existing DT	S-EMP implementation	Log-book	Daily
		Air Quality / Fugitive Emissions				
Increase in air emissions (including fugitive dust) due to movement of heavy equipment, machinery, construction vehicles, civil works (levelling, grading, back-filling, concrete / cement mixing, etc), stockpiles, etc.	E.1	Periodic watering (water spraying) before movement of vehicles, at work sites, open and exposed areas and access roads, temp access tracks, exposed stockpiles (if any) along DT ROW, during concrete / cement mixing process	new DT + existing DT ROW	spraying of water	Ocular inspection / log-book	Daily during peak construction in dry season
	E.2	Prohibit the use of chemical dust suppressants	new DT + existing DT ROW	chemical dust suppressants avoided	Ocular inspection / spot-checks	Frequent spot checks

⁴¹ <https://www.ifc.org/en/insights-reports/2000/general-environmental-health-and-safety-guidelines>; <https://www.ifc.org/content/dam/ifc/doc/2000/2007-general-ehs-guidelines-wastewater-and-ambient-water-quality-en.pdf>; <https://www.ifc.org/content/dam/ifc/doc/2000/2007-electric-transmission-distribution-ehs-guidelines-en.pdf>

Table 4. Construction Stage: New Distribution Lines (DT) + System Strengthening and Reconductoring of Existing Distribution Lines (DT)						
Key Activities / Potential Impacts & Issues	No.	Mitigation Measures and Safeguards	Monitoring Location ³⁹	Monitoring Parameter	Method of Measurement	Frequency
	E.3	Dust prone materials / stockpiles to be stored in enclosures and/or covered with tarp	MSEDCL warehouse / filter unit or Contractor owned warehouses	Materials / Stockpiles storage area	Ocular inspection / spot-checks	Frequent spot checks
	E.4	Periodic maintenance of heavy equipment, machinery, construction vehicles to minimise air pollution, noise, and reduce OHS risks	area of influence	heavy equipment, machinery, construction vehicles	Ocular inspection / Maintenance log-book	Weekly
	E.5	Control vehicle speed to ≤ 8 km/h in unpaved areas including unpaved approach roads; Post the speed limit sign in the project work areas	area of influence	Speed limit signage	Ocular inspection / spot checks	Frequent spot checks
	E.6	Vehicles transporting materials that generate dusts will be covered with tarps	area of influence	Provision of tarps on vehicles	Ocular inspection / spot checks	Frequent spot checks
	E.7	Undertake air quality monitoring using handheld / portable air monitoring devices at select locations for OHS / CHS purposes	only if there are any sensitive receptors downwind of the pole foundation area	PM10, PM2.5, SO _x , NO _x , CO	Monitoring conducted using hand held / portable air monitoring devices	Monthly during construction
	E.8	Comply with relevant National standards on air quality / IFC (WBG) Environmental, Health, and Safety (EHS) Guidelines on Air and Ambient Air Quality, Electric Transmission and Distribution ⁴²	new DT + existing DT ROW	Compliance with National standards and/or IFC (WBG) Guidelines, whichever is stringent	Record and maintain log of monitoring / incidences of non-compliance and rectification	Monthly during construction + once after completion of construction

⁴² <https://www.ifc.org/content/dam/ifc/doc/2000/2007-general-ehs-guidelines-air-emissions-and-ambient-air-quality-en.pdf>; <https://www.ifc.org/content/dam/ifc/doc/2000/2007-electric-transmission-distribution-ehs-guidelines-en.pdf>

Table 4. Construction Stage: New Distribution Lines (DT) + System Strengthening and Reconductoring of Existing Distribution Lines (DT)						
Key Activities / Potential Impacts & Issues	No.	Mitigation Measures and Safeguards	Monitoring Location ³⁹	Monitoring Parameter	Method of Measurement	Frequency
	E.9	Implement S-EMP: Dust Control	new DT + existing DT ROW	S-EMP implementation	Log-book	Daily
		Noise				
Increase in noise due to movement of heavy equipment, machinery, construction vehicles, civil works (levelling, grading, back-filling, etc)	F.1	Construction activities utilizing heavy equipment, machinery work will be restricted between 6 AM – 7 PM; no night time works	new DT + existing DT ROW	Construction timings	Log-book	Daily during construction
	F.2	Advance warning and sharing of Construction Implementation schedule / reconductoring work to nearby households / settlements ⁴³	new DT + existing DT ROW	Consultations, Information dissemination via leaflets	Consultations log-book; leaflets distributed	Once prior to scheduled project activity
	F.3	Ensure all generators, heavy equipment, machinery, etc are sound insulated and placed in enclosures to minimise ambient noise levels; situated away from construction (workers) camps	new DT + existing DT ROW	Guaranteed noise level of corresponding equipment, machinery	Equipment/ machinery specifications	Once prior to scheduled project activity
	F.4	Periodic maintenance of heavy equipment, machinery, construction vehicles to minimise emissions and noise, and reduce OHS risks	area of influence	Construction vehicles, machinery, equipment	Ocular inspection / Maintenance log-book	Weekly
	F.5	Undertake noise quality monitoring using handheld / portable noise monitoring devices at select locations for OHS / CHS purposes	only if there are any sensitive receptors downwind of the work sites	Day time noise	Monitoring conducted using hand held / portable devices	Monthly during peak construction

⁴³ Measures in F.2, Table 4 to be undertaken in coordination with the PIU

Table 4. Construction Stage: New Distribution Lines (DT) + System Strengthening and Reconductoring of Existing Distribution Lines (DT)						
Key Activities / Potential Impacts & Issues	No.	Mitigation Measures and Safeguards	Monitoring Location ³⁹	Monitoring Parameter	Method of Measurement	Frequency
	F.6	Comply with relevant National standards on noise, / IFC (WBG) Environmental, Health, and Safety (EHS) Guidelines on Noise, Electric Transmission and Distribution ⁴⁴	new DT + existing DT ROW	Compliance with National standards and/or IFC (WBG) Guidelines, whichever is stringent	Record and maintain log of monitoring / incidences of non-compliance and rectification	Monthly during peak construction
	F.7	Implement S-EMP: Noise Control	new DT + existing DT ROW	S-EMP implementation	Log-book	Daily
		Polluting Materials and Pollution Management				
Pollution due to oil, lubricant, fuel spillage	G.1	Allocate and maintain a temporary secured and/or controlled area s for storage and handling of hazardous materials, if any, with a containment tray or bunds; away from direct sunlight and rain	new DT + existing DT ROW	Temporary secure and/or controlled area set up in place	Ocular inspection / spot checks	Weekly
Inappropriate handling, storage, transport and disposal	G.2	Collection of oily rags and waste oil in proper containment units	new DT + existing DT ROW	Containment units provided in secured / controlled area	Ocular inspection	Weekly or as required during project activity
	G.3	Record of incidents, spills to be maintained at work sites	new DT + existing DT	Incidents, spills	Ocular inspection / log-book	Daily
	G.4	MSEDCL / licensed vendors to collect used equipment, capacitors, etc for	new DT + existing DT	Vendor list	Log-book with record of pick up	Weekly or as required

⁴⁴ <https://www.ifc.org/content/dam/ifc/doc/2000/2007-general-ehs-guidelines-noise-en.pdf>; <https://www.ifc.org/content/dam/ifc/doc/2000/2007-electric-transmission-distribution-ehs-guidelines-en.pdf>

Table 4. Construction Stage: New Distribution Lines (DT) + System Strengthening and Reconductoring of Existing Distribution Lines (DT)						
Key Activities / Potential Impacts & Issues	No.	Mitigation Measures and Safeguards	Monitoring Location ³⁹	Monitoring Parameter	Method of Measurement	Frequency
		repair, re-use ⁴⁵				
Materials Safety Data Sheet (MSDS)	G.5	Delivery and acceptance of all hazardous materials, equipment to be accompanied by a Materials Safety Data Sheet (MSDS) and /or to be certified that it is Polychlorinated Bi-phenyl (PCB), Chlorofluorocarbons (CFCs) including halon, free	new DT + existing DT	Record of equipment and corresponding PCB, CFCs, Halon free certificate; Record of MSDS	Log-book, record of MSDS onsite, certification checks	Weekly or as required during project activity
Use of Licensed Vendors	G.6	MSEDCL / licensed vendors to collect used cables, capacitors, etc from existing DT for repair, re-use, etc ⁴⁶	Existing DT	Licensed vendor list	Log-book with record of trip, location and distance	As required during reconductoring works
	G.7	Record of materials with corresponding location and distance for final storage / disposal / re-use from work sites with date, location maps and number of trips to be maintained onsite	new DT + existing DT	Materials inventory	Ocular inspection / log-book	Daily
	G.8	Comply with relevant National standards in hazardous and polluting materials / IFC (WBG) Environmental, Health and Safety (EHS) Guidelines for Hazardous Material Management, Waste Management, Electric Transmission and Distribution ⁴⁷	new DT + existing DT	Compliance with National standards and/or IFC (WBG) Guidelines, whichever is stringent	Record and maintain log of monitoring / incidences of non-compliance and rectification	Monthly

⁴⁵ Note: PCB-containing equipment can be used until December 31, 2025 or the equipment's certified lifetime, whichever is earlier. The equipment must be properly maintained to prevent PCBs from leaking or being released into the environment. PCB-contaminated equipment must be regulated as hazardous waste under the Hazardous Wastes (Management, Handling and Transboundary Movement) Rules, 2008.

⁴⁶ Note: PCB-containing equipment can be used until December 31, 2025 or the equipment's certified lifetime, whichever is earlier. The removed equipment shall be properly maintained to prevent PCBs from leaking or being released into the environment. PCB-contaminated equipment is regulated as hazardous waste under the Hazardous Wastes (Management, Handling and Transboundary Movement) Rules, 2008.

⁴⁷ <https://www.ifc.org/content/dam/ifc/doc/2000/2007-general-ehs-guidelines-hazardous-materials-management-en.pdf>; <https://www.ifc.org/content/dam/ifc/doc/2000/2007-general-ehs-guidelines-waste-management-en.pdf>; <https://www.ifc.org/content/dam/ifc/doc/2000/2007-electric-transmission-distribution-ehs-guidelines-en.pdf>

Table 4. Construction Stage: New Distribution Lines (DT) + System Strengthening and Reconductoring of Existing Distribution Lines (DT)						
Key Activities / Potential Impacts & Issues	No.	Mitigation Measures and Safeguards	Monitoring Location ³⁹	Monitoring Parameter	Method of Measurement	Frequency
	G.9	Implement S-EMP: Hazardous Materials Control Plan (including spills response and clean-up)	new DT + existing DT	S-EMP implementation	Log-book	Daily
		Waste Management (all waste streams)				
Improper waste management, handling and disposal	H.1	Prohibit waste burning	new DT + existing DT ROW	Waste management	Ocular inspection / spot checks	Daily
	H.2	Transport of recyclables to dedicated MSEDCL yards / filter units for re-use, repair; resale or auction to government authorized vendors only	new DT + existing DT	Licensed vendor list for auction	Inventory / Log-book with record of trip, location and distance	Immediately after pole erection, construction
	H.3	Ensure proper removal and disposal of any significant residual materials, wastes that remain onsite after completion of civil works; no final disposal on site; No disposal on farmland	new DT + existing DT	Waste management	Record of waste disposal sites with corresponding distance from work site, date, location map and number of trips to be maintained at work site	Once after completion of construction
	H.4	Comply with relevant National standards on waste management / IFC (WBG) Environmental, Health and Safety (EHS) Guidelines on (General and Hazardous) Waste Management, Electric Transmission and Distribution) ⁴⁸	new DT + existing DT	Compliance with National standards and/or IFC (WBG) Guidelines, whichever is stringent	Record and maintain log of monitoring / incidences of non-compliance and rectification	Monthly
	H.5	Implement S-EMP: Waste Management (all waste streams)	new DT + existing DT	S-EMP implementation	Log-book	Daily
		Occupational Health and Safety (OHS)				

⁴⁸ <https://www.ifc.org/content/dam/ifc/doc/2000/2007-general-ehs-guidelines-waste-management-en.pdf>; <https://www.ifc.org/content/dam/ifc/doc/2000/2007-electric-transmission-distribution-ehs-guidelines-en.pdf>

Table 4. Construction Stage: New Distribution Lines (DT) + System Strengthening and Reconductoring of Existing Distribution Lines (DT)						
Key Activities / Potential Impacts & Issues	No.	Mitigation Measures and Safeguards	Monitoring Location ³⁹	Monitoring Parameter	Method of Measurement	Frequency
Provision of constr (workers) camps	I.1	Ensure adequate workers accommodation in line with IFC (WBG) Guidance Note / Workers Accommodation and measures in A.3 - A.5, Table 2	new DT	Provision of accommodation and adequate services	Ocular inspection	Once
Unhygienic conditions at constr (workers) camps	I.2	Construction (workers) camps to have adequate facilities for temporary stay such as electricity, portable water, portable toilets, waste collection areas /bins, first aid kit, etc	new DT	Provision of adequate water and sanitation facilities, electricity connection	Ocular inspection / spot-checks	Weekly
Exposure to hazards for workers working on concrete mixing, cement, etc	I.3	Concrete, cement mixing process for pole foundation located at least 10 m downwind of nearest household / settlement and/or construction (workers) camp	new DT + existing DT	10 m distance maintained	Ocular inspection	Once
Exposure to hazards for workers working on concrete mixing, cement, etc	I.4	Strictly enforce the use of PPE as such goggles, rubber gloves, sleeves, blankets, noise reducing mufflers, head lamps, high visibility safety vests with reflective striping, respirators (e.g., working on concrete / cement, etc)	new DT + existing DT	PPE provided	Inventory and, PPE check / ocular inspection	Weekly
	I.5	Provide communication devices to contractor supervisors / C-EHS focal point person	new DT + existing DT	Communication devices provided	Inventory check	Once
Electrical hazards, Working at heights, etc	I.6	Assess safety risks and safety protocols as per MSEDCL requirements and implement at work sites ⁴⁹	new DT + existing DT	Safety risks / protocols assessed for OHS	Safety check	Weekly
	I.7	When testing electrical equipment, all unrelated workers to leave the 'flagged or danger zone' and comply with the relevant MSEDCL electrical safety standards	new DT + existing DT	Safety risks / protocols assessed for OHS	Safety check	During testing of electrical equipment

⁴⁹ [MSEDCL Training and Safety Guidelines](#)

Table 4. Construction Stage: New Distribution Lines (DT) + System Strengthening and Reconductoring of Existing Distribution Lines (DT)						
Key Activities / Potential Impacts & Issues	No.	Mitigation Measures and Safeguards	Monitoring Location ³⁹	Monitoring Parameter	Method of Measurement	Frequency
	I.8	Provide and maintain signage as per MSEDCL standards for electrical safety	new DT + existing DT ROW	Adequate signage provided	Ocular inspection	Once
	I.9	No night time engagement for working at heights, electrical works, and/or during periods of fog and strong wind on the Beau fort Wind Scale ⁵⁰	new DT + existing DT	Construction timings	Log-book	Daily
	I.10	All workers conducting line works shall use Safety Harness, Helmets, etc and have a Safety Certificate issued by the GOI National Safety Council (NSC) Certification ⁵¹ or equivalent	new DT + existing DT	Hiring of trained workers	Safety certificate on file	Once
Fire, explosion and other incidences / accidents	I.11	Periodic training of workers for specific type of work engagement, e.g., OHS, hazardous materials handling, waste disposal, working at heights, electrical works, safety protocols, and Emergency Response Plans (ERPs) including fire and explosion	area of influence	OHS training	OHS training log-book	Once every 3 months
Health incidences, workers safety	I.12	Coordinate with nearest hospital / local clinic in case of emergencies and/or work accidents	area of influence	Clinic / hospital list	Log-book	Once
	I.13	Ensure provision of first aid kit at work sites	area of influence	Provision of first aid kits	Inventory check / spot checks	Once
	I.14	Provide periodic health check-ups (health & fitness) to workers	area of influence	Health check-ups	Record of health assessments	Once every 3 months
	I.15	Undergo orientation in health and safety risks due to transmittable diseases (HIV, AIDS, Covid-19)	area of influence	Worker's orientation in OHS	Record of orientation	Once every 3 months

⁵⁰ The Beaufort Wind Scale is an empirical measure with 12 wind speed classes. Winds above Class V are higher than 10.8 m/sec

⁵¹ <https://nsc.org.in/>

Table 4. Construction Stage: New Distribution Lines (DT) + System Strengthening and Reconductoring of Existing Distribution Lines (DT)						
Key Activities / Potential Impacts & Issues	No.	Mitigation Measures and Safeguards	Monitoring Location ³⁹	Monitoring Parameter	Method of Measurement	Frequency
	I.16	Maintain a record of health assessments, orientation, incidents, spills, accidents, near miss, fatalities in a log-book at work sites	area of influence	Incidents, spills, accidents, near miss, fatalities	Record of incidences in log-book	Once every 3 months
	I.17	Comply with relevant National laws and regulations on Occupational Health and Safety / IFC (WBG) Environmental, Health, and Safety (EHS) Guidelines on Occupational Health and Safety ⁵²	area of influence	Compliance with National laws and regulations and/or IFC (WBG) Guidelines, whichever is stringent	Record and maintain log of monitoring / incidences of non-compliance and rectification	Monthly
	I.18	Implement S-EMP: Occupational Health and Safety Plan	area of influence	S-EMP implementation	Log-book	Daily
	I.19	Implement S-EMP: Emergency Response Plan	area of influence	S-EMP implementation	Log-book	Daily
		Community Health and Safety (CHS)⁵³				
Excessive disturbances to communities due to civil works, increase in vehicular / equipment / machinery traffic	J.1	Issue advance warning to households / settlements viz area Gram Panchayat including erection of signage of planned route, detours, work timing, duration at appropriate places along road and at work sites; deploy workers to direct traffic in case of lane / road closure	Households / settlements in and around new DT + existing DT ROW	Issuance of advance notice, danger and warning signs, workers deployed corresponding to the construction implementation schedule	Advance warning / information disseminated	As per construction implementation schedule
Disruption to utilities	J.2	Confirm with relevant local agency for existing utilities (UG pipes, telecom, etc) prior to commencing civil / reconductoring works	new DT + existing DT ROW	Damage, disadvantage to local communities, businesses, persons	Ocular inspection / spot-checks	As per construction implementation schedule

⁵² <https://www.ifc.org/content/dam/ifc/doc/2000/2007-general-ehs-guidelines-occupational-health-and-safety-en.pdf>

⁵³ Measures listed under Community Health and Safety J., 1 through J.10, Table 4 to be undertaken in coordination with the PIU

Table 4. Construction Stage: New Distribution Lines (DT) + System Strengthening and Reconductoring of Existing Distribution Lines (DT)						
Key Activities / Potential Impacts & Issues	No.	Mitigation Measures and Safeguards	Monitoring Location ³⁹	Monitoring Parameter	Method of Measurement	Frequency
Lack of access to nearby areas	J.3	Maintain access to grazing sites, adjacent properties, access / pedestrian roads or provide nearest alternate route marked on map	new DT + existing DT ROW	Access maintained	Ocular inspection/ spot checks	During construction
Accidents due to access to work sites	J.4	Ensure all excavations / trenches are secure through appropriate barricades, warning cones (e.g., pole foundations, temp access tracks, etc), signage, sufficient lighting	pole footings along new DT + existing DT ROW	Barricade, warning cones installed, lighting installed, danger and warning signs for safety of households / settlements	Ocular inspection/ spot checks	During construction
Temporary traffic management	J.5	When stringing and stretching conductors across roadways, inform the communities in advance, obtain prior approval from relevant government agency, divert traffic as appropriate in consultation with the local traffic police to ensure public safety, employ scaffolding to support conductors and minimize disruption; employ danger signs in appropriate areas	new DT + existing DT	Issuance of advance notice, danger and warning signs, scaffolding, workers deployed for diverting traffic	Ocular inspection/ spot checks	During stringing and stretching of conductors
CHS issues	J.6	Conduct meaningful consultations with households/ settlements to keep them informed on project implementation schedule (and changes if any), in particular those that may result in disruption wrt area access, vehicular movement, utilities, noisy / dust generating activities, night time works, etc; distribute project information leaflet in local language including Posters on Awareness of Electrical Safety ⁵⁴	area of influence	information disclosure, participation and consultation including distribution of project information leaflet, project GRM, etc	Consultation log-book	Monthly during construction + Once at completion of construction

⁵⁴ [Posters on Awareness of Electrical Safety](#)

Table 4. Construction Stage: New Distribution Lines (DT) + System Strengthening and Reconductoring of Existing Distribution Lines (DT)						
Key Activities / Potential Impacts & Issues	No.	Mitigation Measures and Safeguards	Monitoring Location ³⁹	Monitoring Parameter	Method of Measurement	Frequency
	J.7	Inform and consult with households/ settlements on project GRM	area of influence	Grievances received, resolution if any	Maintain a record of grievances raised in a GRM log-book at work site, records to be maintained in line with project GRM	Monthly during construction + Once at completion of construction
	J.8	Comply with National laws and regulations on CHS/ IFC (WBG) Environmental, Health, and Safety (EHS) Guidelines on CHS ⁵⁵	area of influence	Compliance with National laws and regulations and/or IFC (WBG) Guidelines, whichever is stringent	Record and maintain log of monitoring / incidences of non-compliance and rectification	Monthly
	J.9	Implement S-EMP: Community Health and Safety Plan	area of influence	S-EMP implementation	Log-book	Daily
	J.10	Any damage or hindrance, disadvantage to local communities, businesses, persons etc., caused by the premature removal or insufficient replacement of public utilities is subject to full compensation, at full liability of the Contractor who caused the problem	area of influence	Part of contract clause	Relevant local agency consulted; utilities marked on site maps; records maintained	Once after completion of construction
		Physical Cultural Resources Management (PCRs)⁵⁶				

⁵⁵ <https://www.ifc.org/content/dam/ifc/doc/2000/2007-general-ehs-guidelines-community-health-and-safety-en.pdf>

⁵⁶ Measures in K.1, Table 4 to be undertaken in coordination with the PIU

Table 4. Construction Stage: New Distribution Lines (DT) + System Strengthening and Reconductoring of Existing Distribution Lines (DT)						
Key Activities / Potential Impacts & Issues	No.	Mitigation Measures and Safeguards	Monitoring Location ³⁹	Monitoring Parameter	Method of Measurement	Frequency
Chance Find of PCRs	K.1	Implement S-EMP: Chance Find Procedures for PCRs as per measures K.1. through K.1.4, Table 4 below:	in and around new DT + existing DT + wider area of influence	PCRs	Ocular inspection / spot-checks	Once
	K.1.1	Civil works suspended immediately if PCRs encountered and Contractor supervisor to be informed				
	K.1.2	Prohibit destroying, damaging, defacing or concealing PCRs				
	K.1.3	Inform and consult with the district office of Culture and Fine Arts official				
	K.1.4	Civil works to resume only after thorough investigation and permission from district office of Culture and Fine Arts				

LIST OF SERVICES

SR. NO.	SERVICE NAME	ACTIVITY NUMBER	UOM	SAC CODE	REQ. QTY	VERSION	MATERIAL TYPE
1	Installation Commisioning of	PM.OTH.INS TALL	Activity unit	998736	1		null

Required Documents (To be uploaded online)

Sr. No.	NAME	SECTION	ITEM	DESCRIPTION
1	Price Bid Schedule-3	Price Section	Installation Commissioning	Price Bid Schedule-3
2	Price Bid Schedule-1	Price Section	Installation Commissioning	Price Bid Schedule-1
3	Price Bid Schedule-2	Price Section	Installation Commissioning	Price Bid Schedule-2
4	Additional Documents 3	Technical Section	Installation Commissioning	Additional Documents if any
5	Additional Documents 4	Technical Section	Installation Commissioning	Additional Documents if any
6	Additional Documents 2	Technical Section	Installation Commissioning	Additional Documents if any
7	Additional Documents 1	Technical Section	Installation Commissioning	Additional Documents if any
8	Company Details	Technical Section	Installation Commissioning	Incorporation, Memorandum, Article of Association, List of Board Members etc.
9	Electrical Contractor License	Technical Section	Installation Commissioning	Electrical Contractor License
10	Additional Documents 5	Technical Section	Installation Commissioning	Additional Documents if any
11	Registration Certificate	Technical Section	Installation Commissioning	GST, PF, Labour Laws Registration etc.
12	Qualification Documents	Technical Section	Installation Commissioning of	LOA / Work Order Copy, Completion Certificate for meeting the qualification Criteria of Tender etc.
13	Form of Technical Proposal	Technical Section	Installation Commissioning	Form of Technical Proposal
14	Power of Attorney	Technical Section	Installation Commissioning	Power of Attorney for Submission of Bids.
15	Financial Documents	Commercial Section		Audited Annual Accounts, Turnover, Net worth, CA Certificate etc.
16	EMD	Commercial Section		Details of EMD/ Bid Security Paid
17	Additional Documents 3	Commercial Section		Additional Documents if any
18	Additional Documents 2	Commercial Section		Additional Documents if any
19	Additional Documents 1	Commercial Section		Additional Documents if any